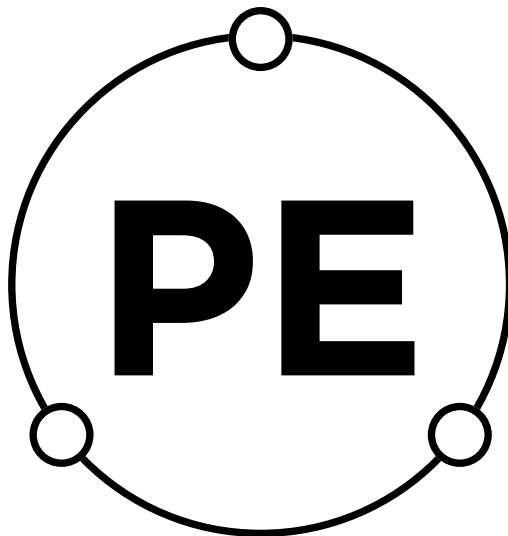


POUNAMU ELECTRONICS
RE_SWC
USER GUIDE

Rev 3.1

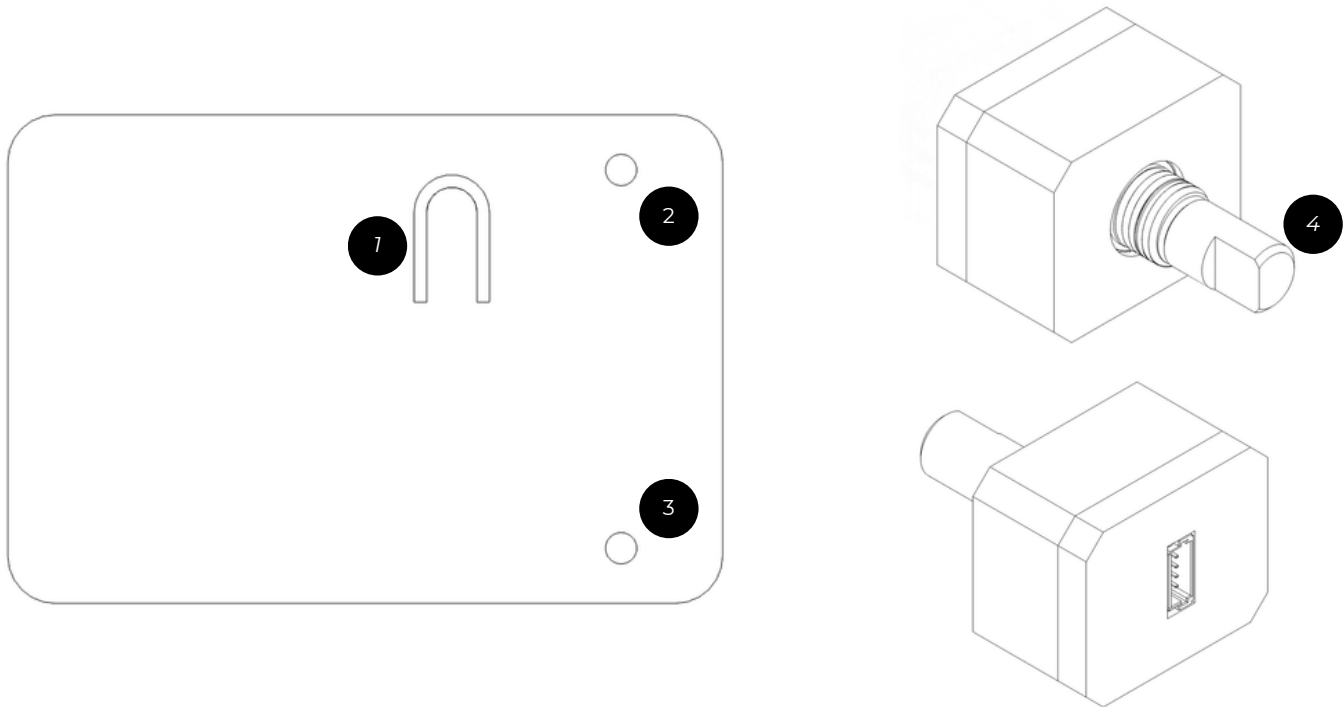


**POUNAMU
ELECTRONICS**

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OVERVIEW



1. UPDATE BUTTON

**2. WHITE STATUS
LIGHT**

**3. RED POWER
LIGHT**

**4. DIA 6mm
ENCODER SHAFT**

ASSEMBLY

USB HID

- Connect the RE_SWC kit to your device with a USB C cable (must support data & **5V power**)
- The RED power indicator on the RE_SWC will illuminate and stay on

Others

- Solder the RED wire from the 2-pin connector on the RE_SWC kit to your headunit's steering wheel control input pin
- Solder the BLACK wire from the 2-pin connector into your harness if your headunit requires a reference for the SWC input (sometimes labelled ST REMOTE GND)
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on

CONFIGURATION

HEADUNIT SELECTION

- *Ensure RE_SWC is not powered*
- *Press and hold the volume knob. Connect power to the RE_SWC while keeping the volume knob held*
- *Release the volume knob after the WHITE status light is illuminated*
- *Press the volume knob once. The WHITE status light will flash once. Headunit number 1 is now programmed. Repeat this step until the desired headunit number is reached - refer to the headunit legend*
- *When the desired headunit number is entered, press and hold the volume knob. The WHITE status light will flash. Release the volume knob. The RE_SWC will flash x times where x is the headunit number entered*

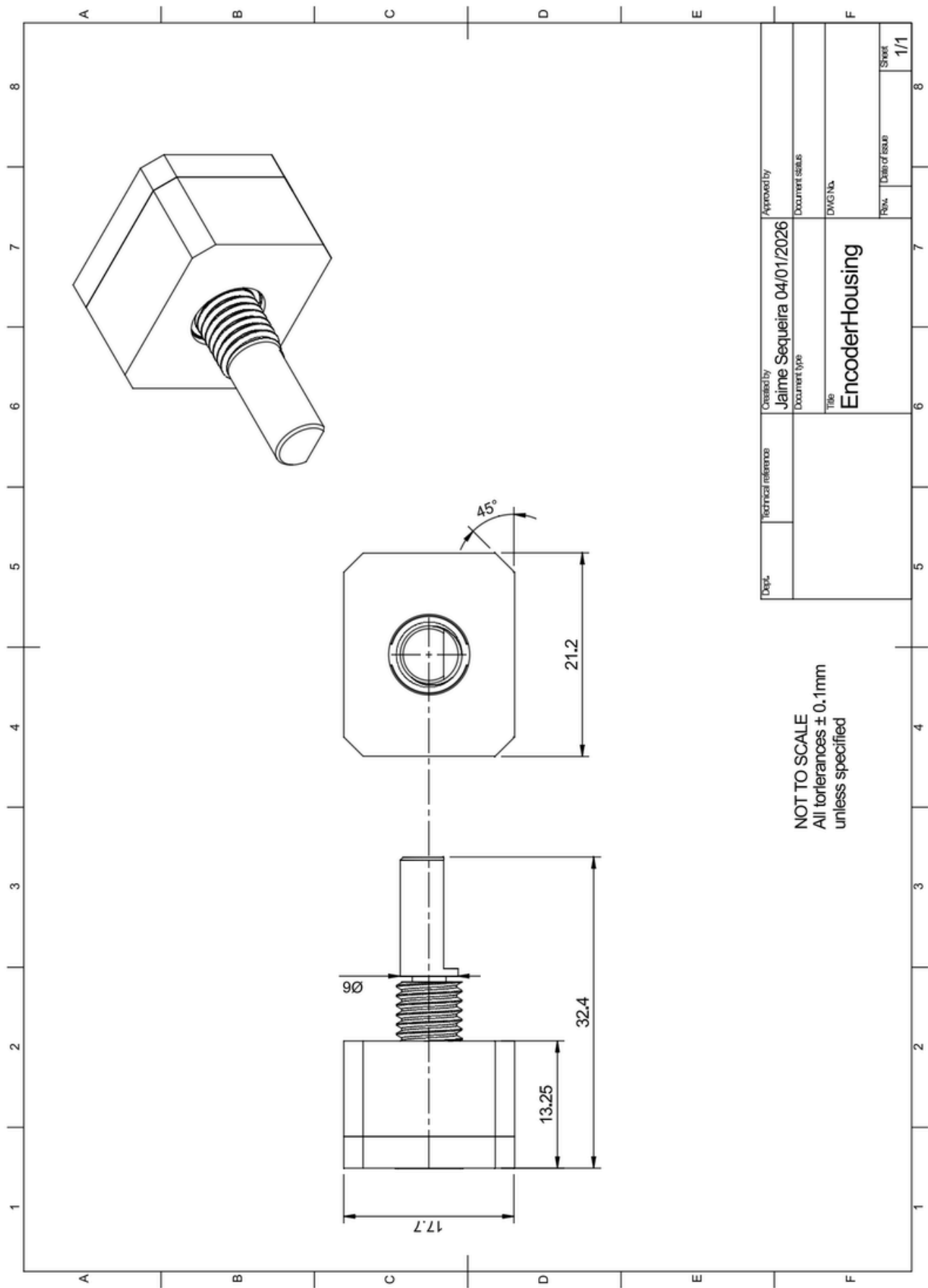
HEADUNIT LEGEND

HEADUNIT NUMBER	HEADUNIT TYPE / BRAND
1	<i>GENERIC RESISTIVE</i>
2	<i>JVC</i>
3	<i>KENWOOD</i>
4	<i>ALPINE</i>
5	<i>PIONEER / SONY</i>
6	<i>USB HID</i>

FUNCTIONS

INPUT	GENERIC RESISTIVE	JVC	KENWOOD	ALPINE	PIONEER / SONY	USB HID
Volume Knob CW	<i>Any</i>	<i>Volume +</i>	<i>Volume +</i>	<i>Volume +</i>	<i>Volume +</i>	<i>Volume +</i>
Volume Knob CCW	<i>Any</i>	<i>Volume -</i>	<i>Volume -</i>	<i>Volume -</i>	<i>Volume -</i>	<i>Volume -</i>
Button Short Press	<i>Any</i>	<i>Mute</i>	<i>Mute</i>	<i>Mute</i>	<i>Mute</i>	<i>Mute</i>
Button Long Press	<i>Any</i>	<i>Next Track</i>	<i>Next Track</i>	<i>Next Track</i>	<i>Next Track</i>	<i>Next Track</i>
Button Double Press	<i>Enter Learning Mode</i>	<i>Previous Track</i>	<i>Previous Track</i>	<i>Previous Track</i>	<i>Previous Track</i>	<i>Previous Track</i>

VOLUME KNOB DIMENSIONS



TROUBLESHOOTING

- **The RED power light is off**
 - *Ensure you are powering the kit with 5V USB power*
- **The RED power light is on but the WHITE status light does not flash when I use the knob**
 - *Ensure the 5-pin connector harness for the knob is seated correctly and inserted fully into the knob and the kit*
- **The WHITE status light flashes when I turn the knob but my headunit does nothing**
 - *Ensure you have set the kit in the correct headunit mode*
 - *Ensure the RED wire from the 2-pin connector is soldered properly into your headunit's harness*
- **My headunit shows the volume increasing but the volume of my music doesn't change in USB HID mode**
 - *Some Android headunits do not route USB volume commands to the correct hardware. Revert to using the Resistive SWC mode with Key1 / Key 2 input wire instead*

RE_SWC SPECS

Input

5V / 100mA USB-C

CONNECTORS

VOLUME KNOB: JST SHR-05V-S-B

SWC OUT: JST SHR-02V-S-B

MCU:

WCH CH32X035F8U6

Digital Rheostat:

MCP4131-104E/SN

SPI @ 250kHz

DISCLAIMERS

- *Pounamu Electronics is not affiliated or associated with any headunit manufacturers or brands*
- *Pounamu Electronics is not affiliated or associated with the USB Implementers Forum (USB-IF)*

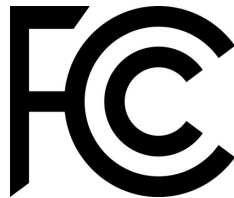
COMPLIANCE

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

- 1. This device may not cause harmful interference, and*
- 2. this device must accept any interference received, including interference that may cause undesired operation.*



Document Revision History

- *Rev 3.1*
 - *Added troubleshooting content*
 - *Edited assembly instructions to include 5V power only spec*
- *Rev 3*
 - *Overhaul for RE_SWC V2*
 - *Added disclaimers section*
 - *Added compliance section*
- *Rev 2 - FW V2.0.0*
 - *Added instructions for JVC headunit interface*
 - *Added instructions for Kenwood headunit interface*
- *Rev 1 - Initial release*